

● HARD

● ADVANCED MATH

● ANSWER KEY

Nonlinear functions

10

2 answers · Rationale-rich · Teach from doc

Q1**B****B. II only**

Choice B is correct. For the function f , since the base of the exponent, 1.25, is greater than 1, the value of 1.25^x increases as x increases. Therefore, the value of 181.25^x and the value of $181.25^x + 41$ also increase as x increases. Since f is therefore an increasing function where $x \geq 0$, the function f has no maximum value. For the function g , since the base of the exponent, 0.73, is less than 1, the value of 0.73^x decreases as x increases. Therefore, the value of 90.73^x also decreases as x increases. It follows that the maximum value of gx for $x \geq 0$ occurs when $x = 0$. Substituting 0 for x in the function g yields $g0 = 90.73^0$, which is equivalent to $g0 = 91$, or $g0 = 9$. Therefore, the maximum value of gx for $x \geq 0$ is 9, which appears as a coefficient in equation II. So, of the two equations given, only II displays, as a constant or coefficient, the maximum value of the function it defines, where $x \geq 0$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q2**35**

The correct answer is 35. It's given that the function $gx = -\frac{1}{55}x + 19x - 35$ gives the depth below the surface of the ocean, in meters, of the submersible device x minutes after collecting a sample, where $x > 0$. It follows that when the submersible device is at the surface of the ocean, the value of gx is 0. Substituting 0 for gx in the equation $gx = -\frac{1}{55}x + 19x - 35$ yields $0 = -\frac{1}{55}x + 19x - 35$. Multiplying both sides of this equation by -55 yields $0 = x + 19x - 35$. Since a product of two factors is equal to 0 if and only if at least one of the factors is 0, either $x + 19 = 0$ or $x - 35 = 0$. Subtracting 19 from both sides of the equation $x + 19 = 0$ yields $x = -19$. Adding 35 to both sides of the equation $x - 35 = 0$ yields $x = 35$. Since $x > 0$, 35 minutes after collecting the sample the submersible device reached the surface of the ocean.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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